

Positioning and Tracking in DECT-2020 NR With Proactive Anchor Selection for Range, Angle, and RSS Measurements

Antti Saikko , *Student Member, IEEE*, Jukka Talvitie , *Member, IEEE*, Joonas Sää , *Member, IEEE*, Juho Pirskanen , and Mikko Valkama , *Fellow, IEEE*

Abstract—This article addresses efficient positioning and user device tracking in Internet-of-Things (IoT) networks with particular focus on the new DECT-2020 New Radio standard—the first noncellular 5G technology standard in the world. Stemming from fundamental performance requirements of IoT networks, for example, related to energy consumption, latency, and reliability, it is important to utilize available radio resources efficiently, while avoiding redundant transmissions and signaling. In this article, we extend our earlier proposed tracking-based positioning solution, which utilized Fisher information to select the most beneficial range and angle measurements, to cover also received signal strength (RSS)-based measurements and particle filter-based solutions. By exploiting prior information inherited from tracking-based positioning solutions, it is possible to proactively select the most valuable positioning measurements, and thus save valuable effort and time in acquiring and processing positioning measurements without sacrificing the positioning performance in practice. Through extensive numerical evaluations, considering range, angle, and RSS measurements, we show that the proposed anchor selection method is able to outperform the traditional signal-to-noise ratio-based measurement selection approach, while enabling positioning with a smaller number of measurements. In addition, we illustrate the effect of prior information quality on the proposed method performance by varying the measurement interval in the tracking process. The numerical results show that when only two anchors are utilized, approximately up to 10%–50% reduction in positioning root-mean-square error can be achieved depending on the considered measurement type.

Index Terms—Digital Enhanced Cordless Telecommunications—2020 New Radio (DECT-2020 NR), extended Kalman filter (EKF), Fisher information, Internet-of-Things (IoT), noncellular 5G, particle filter, positioning, tracking.

I. INTRODUCTION

ACCOMMODATING to growing demand on Internet-of-Things (IoT)-based services, the new Digital Enhanced Cordless Telecommunications—2020 New Radio (DECT-2020 NR) [1], [2] was introduced as part of the official International Telecommunication Union (ITU) fifth generation (5G) standards in 2022 [3]. As the first noncellular 5G technology, the DECT-2020 NR is developed especially for IoT use cases with support of ultrareliable low latency communications (URLLC) and massive machine-type communications (mMTC) related performance requirements from 5G.

In various IoT and wireless sensor network (WSN) systems, along with related applications and location-based services [4], positioning plays an important and emerging role. Except for a few recent studies, [5], [6], DECT-2020 NR-based positioning has not yet been widely studied due to its technological novelty. Nonetheless, research works focusing on positioning with other IoT systems, or IoT-like scenarios, can be found, for example, in [4], [7], [8], [9], [10], [11], and [12]. In [4], different IoT-based

positioning approaches are evaluated and classified in an IoT infrastructure applicable framework, while also considering aspects of power consumption and energy efficiency. Furthermore, potential positioning methods for IoT systems are comprehensively discussed in [7] and [11], including also comparison between positioning capabilities of different IoT technologies.

In this article, we focus on positioning and user device tracking in IoT and WSN like systems using angle-of-arrival (AoA) measurements, time-of-arrival (ToA)-based range measurements and received signal strength (RSS) measurements, obtained from known anchor devices, while specifically focusing on the capabilities and physical-layer of the new DECT-2020 NR standard. Furthermore, we provide and derive fundamental performance bounds, and present Fisher information-based measurement selection method to choose the most important anchor measurements for minimized positioning error. This article extends our early-stage work reported in [13] by introducing utilization of RSS measurements, which are often considered essential to many practical IoT-based positioning systems, and introduces a particle filter-based tracking framework with a novel anchor selection solution. The proposed measurement selection approach has not been earlier presented in the literature, but prior research regarding positioning with range, AoA and RSS measurements have been earlier considered, e.g., in [9], [10], and [14]. Different types of measurement selection approaches with varying selection objectives, not directly related to the focus in this article, have been studied, e.g., in [15], [16], [17], and [18]. In [15], the anchor selection focuses on

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Antti Saikko, Jukka Talvitie, Joonas Sää, and Mikko Valkama are with the Department of Electrical Engineering, Tampere University, 33720 Tampere, Finland (e-mail: antti.saikko@tuni.fi; jukka.talvitie@tuni.fi).

Juho Pirskanen is with wirepas Ltd., 33720 Tampere, Finland.

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nonline-of-sight (NLoS) environments, where NLoS anchors are identified and mitigated by comparing variances of intermediate estimated positions with predefined thresholds. In [16], anchor selection using the Hodges–Lehmann test has been proposed for cooperative positioning framework. In [17], anchor selection is performed based on received signal strength (RSS)-based interval overlap degree method, and in [18] a joint anchor optimization selection through trilateration is utilized. Moreover, a signal-to-noise ratio (SNR)-based approach for measurement and reference anchor selection with time-difference-of-arrival measurements is studied and described in [19], serving as a reference scheme for the results shown in this article.

Through the proposed proactive anchor selection it is possible to reduce the number of required positioning measurements, and subsequently, potentially improve energy efficiency, decrease signaling overhead for increased spectral efficiency, reduce latency, and lower the computational load, which are all essential performance criteria in IoT systems [20]. The proposed measurement selection method utilizes the prior information of an underlying tracking method, such as extended Kalman filter (EKF) or particle filter, and the measurement-associated information from potential anchors to arrange the anchors in order of achievable positioning error. This allows harnessing and utilizing all the information available in the measurements, including the measurement geometry, representing thus a more elaborate approach compared to the SNR-based reference scheme.

The main contributions in this article, incorporating also the contributions of our early-stage research work in [13] are summarized as follows.

- 1) We propose a novel Fisher information-based measurement selection method to choose the most important positioning measurements for minimizing the positioning error, as originally introduced in [13].
- 2) We provide and derive fundamental positioning performance bounds, and propose an EKF-based measurement selection method for practical tracking scenarios, as originally presented in [13].
- 3) We formulate a particle filter-based tracking solution for the proposed measurement selection scheme to manage challenging operation environments with highly nonlinear measurement model behavior.
- 4) We propose a novel Fisher information-based measurement selection method used with particle filters to improve tolerance against large prior uncertainty in anchor selection.
- 5) We extend our system models and the proposed measurement selection schemes in our early-stage research work [13] to cover RSS measurements, and evaluate the performance of RSS-based tracking using a particle filter.
- 6) In comparison with our early-stage research work in [13] we introduce more elaborate discussion and illustrations regarding the proposed selection schemes, and present new results regarding theoretical positioning bounds.

The rest of this article is organized as follows. Section II introduces the DECT-2020 NR with the considered system model. In Section III, theoretical performance bounds based on Fisher information are evaluated for the considered positioning measurements and tracking-based solutions with prior

information. The proposed measurement selection for tracking-based solutions is presented in Section IV, considering both EKF and particle filter-based aspects. Numerical results are presented and discussed in Section V. Finally, Section VI concludes this article.

II. DECT-2020 NR AND SYSTEM MODEL

In this section, we introduce the DECT-2020 NR system, and present the related system parameters. In addition, we define the system model and describe the considered range, angle, and RSS measurements.

A. DECT-2020 NR

DECT-2020 NR is a radio interface technology (RIT) standardized by European Telecommunications Standards Institute, and it is part of International Mobile Telecommunication—2020 (IMT-2020) 5G technologies. Digital Enhanced Cordless Telecommunications - 2020 New Radio (DECT-2020 NR) fills performance requirements for URLLC and mMTC use cases as specified by ITU Radiocommunication Sector for IMT-2020 [3], [21], [22]. The International Mobile Telecommunication (IMT) evaluation process was gone through as a set of radio interface technologies (SRIT) called DECT 5G-SRIT, which consists two RIT components, DECT-2020 NR and 3rd Generation Partnership Project 5G [23]. DECT-2020 NR is potential technology especially for massive IoT use cases and smart IoT applications, such as smart home or smart buildings [1]. However, DECT-2020 NR also supports a wide range of other applications, including cordless telephony, professional audio and audio streaming applications [21], [24]. It is targeted for local area wireless applications, offering easy development and minimal maintenance, while also supporting integration with wide area networks and having self-organizing mesh networking capabilities. DECT-2020 NR supports different types of network topologies, such as point-to-point, point-to-multipoint, and star topologies [21], [23], [25].

The DECT-2020 NR operates at a dedicated license exempt frequency band, available only for digital-enhanced cordless telecommunications (DECT)-based transmissions. Compared to systems operating at nondedicated frequency bands, such as systems using Wi-Fi or Bluetooth-based technologies, the dedicated DECT-band enables reduced overall radio interference. Furthermore, the license exempt operation of DECT-2020 NR has potential for faster technology adoption rate compared to license-based technologies, such as cellular 5G new radio (NR).

The first DECT-2020 NR chipset and protocol implementations have been available from the first quarter of 2024, and now end products are appearing into market. Further information on the DECT-2020 NR and related industry applications and market information can be found in [26]. For example, it is indicated that about 30% of wireless smart electricity meters in 2026 could be addressable market for DECT-2020 NR.

B. Physical-Layer Numerology

The DECT-2020 NR physical layer supports 17 different frequency bands below 6 GHz frequencies [27, table 5.2–1]

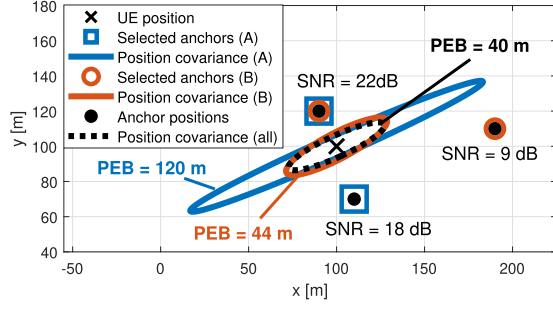


Fig. 1. Illustrative example regarding the effect of anchor selection on PEB in a scenario with three anchors using RSS measurements. Besides selecting all anchors, two separate anchor selection options (A and B) are indicated with different markers and colors.

with a dedicated DECT band located around 1.9 GHz carrier frequency. The specified orthogonal frequency-division multiplexing (OFDM)-based waveform supports a variable subcarrier spacing (SCS), containing 27, 54, 108, and 216 kHz options [28, Table 4.3–1]. Depending on the used SCS, the achievable nominal bandwidth varies from 1.728 MHz (with 27 kHz SCS) to 221.184 MHz (with 216 kHz SCS). Furthermore, multiple-input multiple-output (MIMO) transmissions are supported with 2, 4, and 8 spatial streams [28, Table 6.3.2–1], which also correspond to a minimum number of required antennas and associated digital transmission chains.

The DECT-2020 NR physical layer employs a multiple access scheme that integrates time division duplex with frequency division multiple access and time division multiple access, while the frame duration of 10 ms and slot duration of 0.41667 ms are configured to ensure compatibility with legacy DECT systems [28]. DECT-2020 NR defines physical channels physical control channel and physical data channel, and transmissions are performed using OFDM waveform with a cyclic prefix. Furthermore, subcarriers are modulated based on quadrature amplitude modulation (QAM), incorporating available modulation schemes from binary phase-shift keying to 1024-quadrature amplitude modulation (QAM) [28].

C. Positioning Measurement Model

We consider positioning using AoA, ToA-based range, and RSS measurements obtained from a set of anchors within the signal coverage. Assuming user position $\mathbf{p}[n] = [x[n] \ y[n]]^T$, where $x[n]$ and $y[n]$ are the x - and y -coordinate of the user device at n th time step, the measurement model is defined as

$$\mathbf{z}[n] = \mathbf{h}(\mathbf{p}[n]) + \mathbf{r}[n] \quad (1)$$

where $\mathbf{z}[n] \in \mathbb{R}^N$ represents measurements from N anchors, $\mathbf{h} : \mathbb{R}^2 \rightarrow \mathbb{R}^N$ is a nonlinear measurement model function, and $\mathbf{r}[n] \sim \mathcal{N}(0, \Sigma[n])$ is noise with covariance $\Sigma[n] \in \mathbb{R}^{N \times N}$. Considering a range measurement, the i th element of the measurement model $\mathbf{h}(\mathbf{p}[n])$ can be written as

$$h_{\text{range},i}(\mathbf{p}[n]) = \|\mathbf{p}[n] - \mathbf{p}_{\text{AN},i}[n]\| \quad (2)$$

and correspondingly for an AoA measurement

$$h_{\text{AoA},i}(\mathbf{p}[n]) = \arctan \left(\frac{y[n] - y_{\text{AN},i}[n]}{x[n] - x_{\text{AN},i}[n]} \right) \quad (3)$$

where $\mathbf{p}_{\text{AN},i}[n] = [x_{\text{AN},i}[n] \ y_{\text{AN},i}[n]]^T$ is the known position of the i th measurement anchor, observed at the n th time step. Moreover, considering RSS measurements, the i th element of the measurement model $\mathbf{h}(\mathbf{p}[n])$ is defined according to the log-distance path loss model, similar to models in [29], as

$$h_{\text{RSS},i}(\mathbf{p}[n]) = P_{1\text{m}} - 10\alpha \log_{10} (\|\mathbf{p}[n] - \mathbf{p}_{\text{AN},i}[n]\|) \quad (4)$$

where $P_{1\text{m}}$ is the RSS at 1 m reference distance, and α is the path loss exponent.

III. PERFORMANCE BOUNDS

In this section, we derive the AoA and range estimation bounds and formulate the positioning error bounds for all considered positioning measurements. Finally, we incorporate the effect of user mobility, and present the related posterior positioning bounds by combining tracking-originated *a priori* information with available measurement-based information.

A. AoA and Range Estimation Bounds

Potential accuracy for AoA and range estimation considering DECT-2020 NR system parameters are addressed using Cramér–Rao lower bounds (CRLBs). For antenna array-based AoA estimation, the CRLB for the estimator variance (in radians) can be written as [30]

$$\sigma_{\text{AoA}}^2 \geq \frac{6\lambda^2}{4\gamma\pi^2 L(L^2 - 1)d_{\text{ant}}^2 \cos^2(\theta)} \quad (5)$$

where L is number of array elements, γ is the SNR, d_{ant} is the antenna element spacing, and θ is the true AoA.

Considering OFDM waveform with uniform subcarrier power allocation, the CRLB for ToA-based ranging can be expressed as [30]

$$\sigma_{\text{range}}^2 \geq \frac{3c^2}{8\gamma\pi^2 f_{\text{SC}}^2 M(M+1)(2M+1)} \quad (6)$$

where f_{SC} is subcarrier spacing. In addition, M is the number of subcarriers at one side of the spectrum, excluding the subcarrier zero so that the overall number of subcarriers is $2M+1$, thus resulting in bandwidth of $B_T = f_{\text{SC}}(2M+1)$.

B. Fisher Information and Measurement-Based PEB

A theoretical lower bound for the covariance matrix of an unbiased estimator $\hat{\mathbf{p}}[n]$ can be formally written as [31]

$$\text{CRLB}(\mathbf{p}[n]) = \mathbf{J}^{-1}(\mathbf{p}[n]) \leq \mathbb{E}_{\hat{\mathbf{p}}[n]} \{ \epsilon_{\mathbf{p}} \epsilon_{\mathbf{p}}^T \} \quad (7)$$

where $\mathbf{J}(\mathbf{p}[n]) \in \mathbb{R}^{2 \times 2}$ is the Fisher information matrix (FIM), $\epsilon_{\mathbf{p}} = \hat{\mathbf{p}}[n] - \mathbf{p}[n]$, and $\mathbb{E}_{\hat{\mathbf{p}}[n]} \{ \cdot \}$ denotes the expectation with respect to $\hat{\mathbf{p}}[n]$. Based on the FIM, the positioning error bound (PEB) can be expressed as [32]

$$\text{PEB}(\mathbf{p}[n]) = \sqrt{\text{trace}(\mathbf{J}^{-1}(\mathbf{p}[n]))}$$

$$= \sqrt{\text{CRLB}(x[n]) + \text{CRLB}(y[n])} \quad (8)$$

where the PEB represents the lower bound of the positioning root-mean-square error (RMSE).

Fisher information for a set of measurements $\mathbf{z}[n]$, such as in (1), is described through a measurement-based FIM $\mathbf{J}_{\text{meas}}(\mathbf{p}[n]) \in \mathbb{R}^{2 \times 2}$, whose element at the i th row and j th column is written as

$$[\mathbf{J}_{\text{meas}}(\mathbf{p}[n])]_{i,j} = -\mathbb{E}_{\mathbf{p}[n]} \left\{ \frac{\partial^2 \ell(\mathbf{z}[n]|\mathbf{p}[n])}{\partial p_i[n] \partial p_j[n]} \right\} \quad (9)$$

where $\ell(\mathbf{z}[n]|\mathbf{p}[n])$ denotes the log-likelihood function of the measurements given the user position $\mathbf{p}[n]$, and $p_i[n]$ is the i th element of $\mathbf{p}[n]$. Furthermore, by assuming the Gaussian measurement model in (1), the measurement-based FIM can be conveniently written as [31]

$$\mathbf{J}_{\text{meas}}(\mathbf{p}[n]) = \mathbf{H}(\mathbf{p}[n])^T \mathbf{\Sigma}^{-1}[n] \mathbf{H}(\mathbf{p}[n]) \quad (10)$$

where $\mathbf{H}(\mathbf{p}[n]) \in \mathbb{R}^{N \times 2}$ is the Jacobian matrix of the measurement function $\mathbf{h}(\mathbf{p}[n])$, defined with respect to $\mathbf{p}[n]$. With uncorrelated position measurements, the measurement covariance matrix is diagonal, and $\mathbf{\Sigma} = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_N^2)$. Accordingly, the measurement-based FIM can be presented as a sum of individual measurement FIMs as

$$\begin{aligned} \mathbf{J}_{\text{meas}}(\mathbf{p}[n]) &= \sum_{i=1}^N \mathbf{J}_{\text{meas},i}(\mathbf{p}[n]) \\ &= \sum_{i=1}^N \frac{1}{\sigma_i^2[n]} \mathbf{H}_i(\mathbf{p}[n])^T \mathbf{H}_i(\mathbf{p}[n]) \end{aligned} \quad (11)$$

where $\mathbf{J}_{\text{meas},i}(\mathbf{p}[n])$ and $\sigma_i^2[n]$ denote the Fisher information and variance of the i th measurement, respectively. Moreover, $\mathbf{H}_i(\mathbf{p}[n]) \in \mathbb{R}^{1 \times 2}$ is the i th row of $\mathbf{H}(\mathbf{p}[n])$, represented for the range, angle, and RSS measurements in (2), (3), and (4), as

$$\begin{aligned} \mathbf{H}_{\text{range},i}(\mathbf{p}[n]) &= \begin{bmatrix} \tilde{x}_i[n] & \tilde{y}_i[n] \\ d_i[n] & d_i[n] \end{bmatrix} \\ \mathbf{H}_{\text{AoA},i}(\mathbf{p}[n]) &= \begin{bmatrix} -\tilde{y}_i[n] & \tilde{x}_i[n] \\ d_i^2[n] & d_i^2[n] \end{bmatrix} \\ \mathbf{H}_{\text{RSS},i}(\mathbf{p}[n]) &= \begin{bmatrix} -10 \alpha \tilde{x}_i[n] & -10 \alpha \tilde{y}_i[n] \\ d_i^2[n] \log(10) & d_i^2[n] \log(10) \end{bmatrix} \end{aligned} \quad (12)$$

in respective order, where $\tilde{x}_i[n] = x[n] - x_{\text{AN},i}[n]$, $\tilde{y}_i[n] = y[n] - y_{\text{AN},i}[n]$, and $d_i[n] = \|\mathbf{p}[n] - \mathbf{p}_{\text{AN},i}[n]\|$.

An example considering the effect of measurement selection on PEB is illustrated in Fig. 1, where RSS measurements from three potential anchors are available for user positioning, while assuming the urban micropath loss model in [29, Table 7.4.1–1]. The figure shows the 25% confidence ellipses and related PEBs for three separate anchor selection options (selecting all anchors, or two anchors with options A and B), as well as the anchorwise SNRs. When using all three anchors the PEB is about 40 m, whereas when selecting only the two leftmost anchors (option A) or the two topmost anchors (option B), the PEB is 120 or 44 m, respectively. Obviously, when using only two anchors, the anchor selection has a great effect on the PEB, as with one option the PEB is comparable to using all anchors, and with other option the PEB collapses to almost three times larger error. In

addition, it is noted that in this example the two anchors with the highest SNR are identical to the ones found in the selection option A. Thus, the traditional SNR-based selection would result in the worst performance with 120 m PEB.

However, SNR-based selection does not generally result in the worst performance, but it is dependent on the measurement geometry and related anchorwise SNR levels. Although in some scenarios the SNR-based measurement selection might lead to the minimum attainable PEB, the proposed Fisher information-based selection method inherently considers both the measurement geometry and associated SNR levels, and has capability to reach the best possible positioning accuracy in all scenarios.

C. User Mobility and Posterior PEB

Considering measurement time series, temporal correlation of user position can be exploited to enhance positioning performance through appropriate tracking algorithms. To this end, we define a user state vector at a given time step n as

$$\mathbf{s}[n] = [x[n], y[n], v_x[n], v_y[n]]^T \in \mathbb{R}^4 \quad (13)$$

where $x[n]$ and $y[n]$ are the x - and y -coordinate of the user device, and $v_x[n]$ and $v_y[n]$ are the device velocities in the x - and y -direction, respectively.

For mobility modeling, we consider a linear state-transition model between consecutive time-instants so that

$$\mathbf{s}[n] = \mathbf{F}\mathbf{s}[n-1] + \mathbf{q}[n], \text{ where} \quad (14)$$

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_{2 \times 2} & \Delta t \cdot \mathbf{I}_{2 \times 2} \\ \mathbf{0}_{2 \times 2} & \mathbf{I}_{2 \times 2} \end{bmatrix} \in \mathbb{R}^{4 \times 4} \quad (15)$$

is the state-transition matrix, $\mathbf{q}[n] \sim \mathcal{N}(0, \mathbf{Q}[n])$ denotes process noise, and Δt defines the time interval between the consecutive time instants. Furthermore, by assuming a constant white-noise acceleration (CWNA)-based mobility model [33], the process noise covariance matrix is given as

$$\mathbf{Q} = \sigma_a^2 \begin{bmatrix} \frac{\Delta t^3}{3} \mathbf{I}_{2 \times 2} & \frac{\Delta t^2}{2} \mathbf{I}_{2 \times 2} \\ \frac{\Delta t^2}{2} \mathbf{I}_{2 \times 2} & \Delta t \mathbf{I}_{2 \times 2} \end{bmatrix} \quad (16)$$

where σ_a^2 is the power spectral density of device acceleration.

With a tracking-based estimator the mobility model in (14) conveys *a priori* information on top of measurement information as [34]

$$\mathcal{J}_{\text{post}}(\mathbf{s}[n]) = \mathcal{J}_{\text{prior}}(\mathbf{s}[n]) + \mathcal{J}_{\text{meas}}(\mathbf{s}[n]) \quad (17)$$

where $\mathcal{J}_{\text{post}}(\mathbf{s}[n])$ is the FIM of the a posteriori estimate, $\mathcal{J}_{\text{prior}}(\mathbf{s}[n])$ is the FIM of the *a priori* estimate, and $\mathcal{J}_{\text{meas}}(\mathbf{s}[n]) = [\mathbf{J}_{\text{meas}}(\mathbf{p}[n]) \quad \mathbf{0}_{2 \times 2}]$ is measurement-based FIM according to (10). Moreover, the prior information of the state can be formally described as

$$[\mathcal{J}_{\text{prior}}(\mathbf{s}[n])]_{i,j} = -\mathbb{E}_{\mathbf{s}} \left\{ \frac{\partial^2 \log \mathcal{P}(\mathbf{s}[n])}{\partial s_i[n] \partial s_j[n]} \right\} \quad (18)$$

where $\mathcal{P}(\mathbf{s}[n])$ denotes the prior probability distribution of state $\mathbf{s}[n]$. However, considering the mobility model in (14), the *a priori* FIM can be derived based on the a posteriori estimate

Algorithm 1: Generic FIM-Based Measurement Selection.

Input: Potential measurement-wise FIMs $\mathcal{J}_{\text{meas},i} = \text{blkdiag}(\mathbf{J}_{\text{meas},i}(\mathbf{s}^-[n]), \mathbf{0}_{2 \times 2})$ for $i \in \Omega_{\text{potential}}$, a priori FIM $\mathcal{J}_{\text{prior}} = (\hat{\mathbf{P}}^-[n])^{-1}$, and the number of measurements to be chosen N_{chosen} .¹

Output: Set of chosen measurement indices Ω_{chosen} .

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1: Initialize with empty set:  $\Omega_{\text{chosen}} = \emptyset$ 
2: while  $|\Omega_{\text{chosen}}| \leq N_{\text{chosen}}$  do
3:   Compute Fisher information for  $i \notin \Omega_{\text{chosen}}$ :
4:    $\mathcal{J}_{\text{post},i} \leftarrow [\mathcal{J}_{\text{prior}} + \sum_{k \in \Omega} \mathcal{J}_{\text{meas},k}] + \mathcal{J}_{\text{meas},i}$ 
5:   Compute squared a posteriori PEB:2
6:    $\sigma_{\text{PEB},i}^2 \leftarrow \text{trace}([\mathcal{J}_{\text{post},i}]_{1:2,1:2}^{-1})$ 
7:   Choose measurement with the minimum PEB:
8:    $\hat{i} \leftarrow \arg \min_i \{\sigma_{\text{PEB},i}^2\}$ 
9:   Add  $\hat{i}$  to the chosen measurement set:
10:   $\Omega_{\text{chosen}} \leftarrow \Omega_{\text{chosen}} \cup \{\hat{i}\}$ 
11: end while

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of the previous time step as

$$\mathcal{J}_{\text{prior}}(\mathbf{s}[n]) = (\mathbf{F}\mathcal{J}_{\text{post}}^{-1}(\mathbf{s}[n-1])\mathbf{F}^T + \mathbf{Q}[n])^{-1}. \quad (19)$$

In practical evaluations, *a priori* FIM of the first time step can be initialized as a diagonal matrix with near zero diagonal elements, corresponding to large state uncertainty. The bounds will be numerically evaluated in Section V while also serve as inputs to the proposed measurement selection methods, described next.

IV. PROPOSED METHODS

In this section, we describe the proposed measurement selection approach, which utilizes *a priori* information from the considered tracking approaches, including an EKF and a particle filter. Also, we propose a supplementary measurement selection scheme for particle filters to increase tolerance against large prior uncertainties in measurement selection.

A. EKF-Based Tracking and Extraction of Prior Information

EKF-based tracking can be utilized for practical position estimation by iterating between a prediction step and an update step, while exploiting the measurement model and mobility model in (1) and (14), respectively [35]. In the prediction step at time step n , the mean and covariance of the *a priori* estimate of the state $\mathbf{s}[n]$ are obtained as

$$\begin{aligned} \hat{\mathbf{s}}^-[n] &= \mathbf{F}\hat{\mathbf{s}}^+[n-1] \in \mathbb{R}^4 \\ \hat{\mathbf{P}}^-[n] &= \mathbf{F}\hat{\mathbf{P}}^+[n-1]\mathbf{F}^T + \mathbf{Q}[n] \in \mathbb{R}^{4 \times 4} \end{aligned} \quad (20)$$

respectively, where $\hat{\mathbf{s}}^+[n-1] \in \mathbb{R}^4$ and $\hat{\mathbf{P}}^+[n-1] \in \mathbb{R}^{4 \times 4}$ are the mean and covariance of the *a posteriori* estimate of the

¹ Besides the state vector definition in (13), this formulation enables utilization of an arbitrarily chosen state vector.

² In addition to, or instead of, the positioning error bound (PEB), other tracked state parameters can be considered in this metric, such as velocity, acceleration, and clock bias.

previous time step. In this article, the *prediction step* is in a fundamental role in the proposed measurement selection approach, which utilizes available prior information at each time step. By observing the analogy between the prior information in (19) and the predicted state covariance in (20), estimated prior information at time instant n can be obtained through the inverse of the predicted state covariance as [34]

$$\hat{\mathcal{J}}_{\text{prior}}(\hat{\mathbf{s}}^-[n]) = (\hat{\mathbf{P}}^-[n])^{-1}. \quad (21)$$

After the prediction step, by using a set of obtained measurements $\mathbf{z}[n]$ the *a posteriori* estimates of the mean and covariance can be acquired with the update step as

$$\begin{aligned} \hat{\mathbf{s}}^+[n] &= \hat{\mathbf{s}}^-[n] + \mathbf{K}[n](\mathbf{z}[n] - \mathbf{h}(\hat{\mathbf{s}}^-[n]_{1:2})) \\ \hat{\mathbf{P}}^+[n] &= (\mathbf{I}_4 - \mathbf{K}[n]\mathbf{H}[n])\hat{\mathbf{P}}^-[n] \end{aligned} \quad (22)$$

respectively, where $\mathbf{K}[n] = \hat{\mathbf{P}}^-[n]\mathbf{H}[n]^T(\mathbf{H}[n]\hat{\mathbf{P}}^-[n]\mathbf{H}[n]^T + \Sigma[n])^{-1} \in \mathbb{R}^{4 \times N}$ is the Kalman gain.

B. Extension to Particle Filter-Based Tracking in Challenging Scenarios With Nonlinear Behavior

In scenarios with highly nonlinear model behavior, EKF-based tracking performance can quickly deteriorate and become unstable with increased risk of diverging. For example, in this study such behavior was encountered with EKF-based tracking when using RSS measurements. In such scenarios with substantial nonlinear behavior it is possible to utilize, for example, unscented Kalman filter (UKF), cubature Kalman filter (CKF), or particle filter, from which in this article we have considered the latter.

With particle filters [36], [37], the nonlinear posterior density at time step n is represented using a set of N_p particles, $\{\mathbf{s}^{(j)}[n], w^{(j)}[n]\}_{j=1}^{N_p}$, as

$$p(\mathbf{s}[n] | \mathbf{z}[n]) \approx \sum_{j=1}^{N_p} w^{(j)}[n] \delta(\mathbf{s}[n] - \mathbf{s}^{(j)}[n]) \quad (23)$$

where $\mathbf{s}^{(j)}[n]$ and $w^{(j)}[n]$ are the state and weight of the j th particle, respectively, and $\sum_{j=1}^{N_p} w^{(j)}[n] = 1$. In this article we utilize sequential importance sampling [36], where a proposal distribution is determined by the state-transition prior based on (14), and thus the j th particle at time instant n is sampled as

$$\mathbf{s}^{(j)}[n-1] \sim p(\mathbf{s}[n] | \mathbf{s}^{(j)}[n-1]). \quad (24)$$

Moreover, assuming independent measurements at different time instants, the particle weights can be updated directly based on the measurement likelihood $w^{(j)}[n] \propto p(\mathbf{z}[n] | \mathbf{s}^{(j)}[n])$. For example, assuming the measurement model in (1) the likelihood is Gaussian with mean $\mathbf{h}(\mathbf{p}[n])$ and covariance $\Sigma[n]$. To manage particle degeneracy, if the estimated particle efficiency $\hat{N}_{\text{eff}} = 1 / \sum_{j=1}^{N_p} (w^{(j)}[n])^2$ at time step n falls below a given threshold value, resampling is performed based on traditional multinomial resampling [36].

Algorithm 2: Simplified Measurement Selection With 2D PEB.

Input: Potential measurement-wise FIMs $\mathbf{J}_{\text{meas},i} = \mathbf{J}_{\text{meas},i}(\hat{\mathbf{s}}^-[n])$ for $i \in \Omega_{\text{potential}}$, a priori FIM $\mathbf{J}_{\text{prior}} = [(\hat{\mathbf{P}}^-[n])^{-1}]_{1:2,1:2}$, and the number of chosen measurements N_{chosen} .

Output: Set of chosen measurement indices Ω_{chosen}

- 1: Initialize with empty set: $\Omega_{\text{chosen}} = \emptyset$
 - 2: **while** $|\Omega_{\text{chosen}}| \leq N_{\text{chosen}}$ **do**
 - 3: Compute for $i \notin \Omega_{\text{chosen}}$:
 - 4: $S_{11,i} \leftarrow [\mathbf{J}_{\text{prior}}]_{1,1} + [\mathbf{J}_{\text{meas},i}]_{1,1} + \sum_{k \in \Omega} [\mathbf{J}_{\text{meas},k}]_{1,1}$
 - 5: $S_{12,i} \leftarrow [\mathbf{J}_{\text{prior}}]_{1,2} + [\mathbf{J}_{\text{meas},i}]_{1,2} + \sum_{k \in \Omega} [\mathbf{J}_{\text{meas},k}]_{1,2}$
 - 6: $S_{22,i} \leftarrow [\mathbf{J}_{\text{prior}}]_{2,2} + [\mathbf{J}_{\text{meas},i}]_{2,2} + \sum_{k \in \Omega} [\mathbf{J}_{\text{meas},k}]_{2,2}$
 - 7: Compute squared a posteriori PEB:
 - 8: $\sigma_{\text{PEB},i}^2 \leftarrow \frac{S_{11,i} + S_{22,i}}{S_{11,i}S_{22,i} - S_{12,i}^2}$
 - 9: Choose measurement with the minimum bound:
 - 10: $\hat{i} \leftarrow \arg \min_i \{\sigma_{\text{PEB},i}^2\}$
 - 11: Add $\hat{i}$ to the chosen measurement set:
 - 12: $\Omega_{\text{chosen}} \leftarrow \Omega_{\text{chosen}} \cup \{\hat{i}\}$
 - 13: **end while**
-

C. Proposed FIM-Based Measurement Selection Using the Predicted a Priori Information

As visible in (11), each potential anchor measurement is capable of accumulating different amount of Fisher information to the positioning task. Consequently, when the number of used anchors is limited, to save energy for instance, it is important to select measurements with the highest information. To this end, the proposed measurement selection method utilizes the prior information provided by an appropriate tracking method, such as the EKF presented in Section IV-A or the particle filter in Section IV-B. As shown in (11) and (12), evaluating the amount information per potential measurement requires knowledge on measurement variances, positions of potential anchors, and the predicted user position at the intended time of measurements. From these, measurement variances are inherent part of standard EKF parameterization and can be estimated, for example, through CRLB evaluations. In addition, anchor positions are assumed to be known as part of the considered positioning framework, while the user position estimate is directly available through the prediction step, for example, as given in (20).

For all time instants n , the first step in the proposed measurement selection method is to evaluate the measurement-associated FIMs for potential anchor measurements, as described in (11). Such set of potential anchors, denoted as $\Omega_{\text{potential}}$, include all anchors within the signal coverage that are able to provide positioning measurements. Nevertheless, minimizing the PEB solely based on the measurement-associated FIMs does not minimize the a posteriori PEB which is also affected by a priori information, as shown in (17). To this end, the second step in the proposed measurement selection method is to evaluate the a priori information by exploiting the predicted state covariance according to (21). Finally, considering a desired number of chosen measurements N_{chosen} , the eventual set of chosen anchors, denoted as Ω_{chosen} with $|\Omega_{\text{chosen}}| = N_{\text{chosen}}$, is

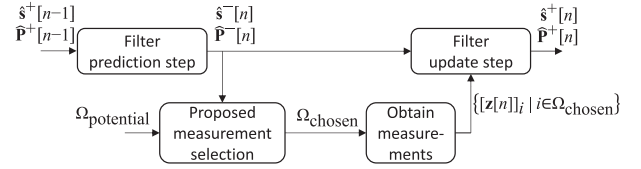


Fig. 2. Illustration of the tracking process including the prediction step, the update step, and the proposed FIM-based measurement selection. The process can be generalized to any predict and update-steps including such with particle filters.

determined by iteratively accumulating measurementwise a posteriori information in accordance with minimum observed PEB. Alternatively, in theory it would be possible to evaluate the a posteriori FIM for all possible measurement combinations. However, in such approach the number of possible measurement combinations grows rapidly with the number of anchors, thus resulting in poor scalability and unfeasible computational burden.

A generic form of the proposed measurement selection method is presented in Algorithm 1. There, besides minimizing PEB, the algorithm can be modified to minimize error with respect to any state vector parameter and enable specific parameterwise error weighting. Nonetheless, when focusing on the 2-D positioning scenario in this article, we can simplify the processing steps by utilizing properties of 2×2 matrix inversion, while working only with position-related submatrices of FIMs. The description of the simplified measurement selection approach with 2-D positioning focus is presented in Algorithm 2. Finally, the overall tracking process utilizing the proposed measurement selection method is illustrated in Fig. 2, where the block of proposed measurement selection involves the processing steps described in Algorithms 1 or 2.

In some cases the covariance of the prior distribution is very large, and the predicted user position does not appropriately reflect the true user position. In such scenarios, to avoid large errors, the proposed FIM-based measurement selection can be adaptively switched OFF, and select measurements according to estimated SNR-values. This type of adaptive scheme can reduce the largest errors considerably, as illustrated later with numerical results in Section V. However, such scenarios with large prior covariance can be alternatively managed with particle filter-based solution, as discussed in the following.

D. Proposed Measurement Selection Using Approximated Prior Distribution of a Particle Filter

When the prior covariance is very large, the mean value of the prior distribution (i.e., the estimated user position), utilized in evaluating the FIMs for the measurement selection, can differ substantially from the true user position. This might result in misleading FIMs, and consequently in unfit measurement selection decisions. To this end, instead of considering only the prior mean, particle filters introduce an inherent capability to examine anchor selection in particlewise manner.

Let us assume a prior distribution represented with particles $\{\mathbf{s}^{(j)}[n], w^{(j)}[n]\}_{j=1}^{N_p}$ and a set of potential anchors $i \in \Omega_{\text{potential}}$ providing positioning measurements. Then, by applying (11) it is possible to evaluate particlewise FIMs for each

potential measurement, denoted as $\mathbf{J}_{\text{meas},i}^{(j)}(\mathbf{p}^{(j)}[n])$, where $\mathbf{p}^{(j)}[n] = [\mathbf{s}^{(j)}[n]]_{1:2}$ is the position of j th particle. Accordingly, a weighted average FIM for the i th potential measurement is

$$\bar{\mathbf{J}}_{\text{meas},i} = \sum_{j=1}^{N_p} w^{(j)}[n] \mathbf{J}_{\text{meas},i}^{(j)}(\mathbf{p}^{(j)}[n]). \quad (25)$$

The weighted average FIM can be used as such in the proposed FIM-based anchor selection, presented in Section IV-C, by replacing $\mathbf{J}_{\text{meas},i}$ with $\bar{\mathbf{J}}_{\text{meas},i}$ in Algorithms 1 or 2.

Instead of evaluating FIMs at a single assumed position of the user (i.e., the prior mean), the weighted average FIM in (25) exploits larger area, and thus incorporate the effect of positioning uncertainty in the proposed anchor selection. Similar approach can be derived also for other tracking schemes, such as Kalman filters, by deriving measurement FIMs at different positions according to the underlying prior covariance. For example, with CKF and UKF a straightforward approach would be utilizing the sigma points from the linearization process. However, particle filters inherently possess a generic form prior distribution with discrete particle positions for evaluating FIMs. It should be noted that it is not required to use all the particles to evaluate the weighted average FIM in (25), but considering only certain particles is feasible to reduce computational complexity.

V. NUMERICAL RESULTS AND ANALYSIS

This section presents numerical evaluations for the considered positioning schemes with the proposed proactive measurement selection. First, the simulation parameters and underlying system assumptions are presented. After this, the theoretical bounds regarding the AoA and range measurements as well as the posterior PEBs are evaluated for different scenarios. Then, evaluations for AoA and range-based methods are presented using EKF-based tracking, followed by corresponding evaluations for RSS measurements using particle filter.

A. Simulation Parameters and Assumptions

For all simulations we consider transmission at the dedicated DECT-2020 NR carrier frequency of $f_c = 1.9$ GHz and SCS of $f_{\text{SC}} = 27$ kHz [1], [38]. Except for ranging accuracy results with variable bandwidth, we fix the number of active subcarriers to 225, corresponding to transmission bandwidth of $B_T = 6.075$ MHz. Regarding AoA estimation, each anchor is equipped with $L = 4$ antennas as the baseline, by using half wavelength antenna spacing $d_{\text{ant}} = c/(2f_c)$, while the cases of $L = 2, 8$ are also considered. The above-mentioned physical layer configurations follow the DECT-2020 NR specifications in [28].

We assume a circular operation area with 690 m radius, where 1000 anchors are randomly deployed based on uniform distribution. The resulted anchor average density of 669 per km² follows roughly the routing device densities reported in [38] and [39]. The AoA- and ToA-based range measurements are simulated as zero-mean Gaussian random variables based on the variances given in (5) and (6), respectively. In addition, we include a random Gaussian clock error with 3 ns standard

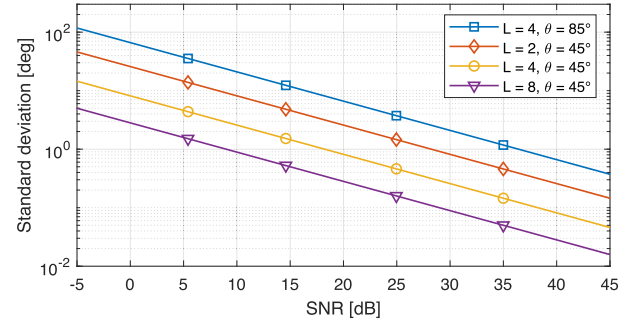


Fig. 3. CRLBs for AoA estimation as a function of SNR with the considered DECT-2020 NR antenna configurations.

deviation to consider synchronization limitations regarding the ToA measurements. The path loss model is defined according to the urban micro path loss model [29, Table 7.4.1–1], while considering only measurements from line-of-sight (LoS) links according to probabilities given in [29, Table 7.4.2–1]. In practice, multipath propagation and inaccuracies in LoS detection can induce outliers in measurements, in which case the outliers can be managed using, for example, the Bayesian information criterion [40]. Considering a need for positioning related signaling between the user device and anchors, SNR of $\gamma > 5$ dB is required for the potential anchors to provide measurements.

The results are evaluated over 100 separate user tracks with ten independent realizations of measurements and anchor deployments. The user device moves smoothly with 1 m/s velocity in the operation area with the average power spectral acceleration of $\sigma_a^2 = 12.2 \cdot 10^{-6} \text{ m}^2/\text{s}^3$. The results are evaluated for different measurement time intervals Δt , including 0.1, 1, and 10 s. Furthermore, the user tracks are not generated using the same mobility model as the one used with the tracking filters. In addition, considering that the proposed tracking schemes utilize the standard CWNA-based mobility model, described in Section III-C, the tracking schemes are expected to support various user mobility scenarios. To enable consistent performance over all scenarios we fix the number of particles in the particle filter as 1000, and perform resampling if the estimated particle efficiency falls below 0.5. The initial user position is estimated using a nonlinear least squares method.

B. Evaluation of Theoretical Bounds

In Fig. 3, based on (5), potential AoA estimation accuracies for the fundamental DECT-2020 NR MIMO configurations, including 2, 4, and 8 antennas, are shown as a function of SNR, while also illustrating the effect of true AoA on the accuracy with $L = 4$. Regarding range estimation, corresponding potential ranging accuracies with the considered DECT-2020 NR numerology are illustrated in Fig. 4 over different bandwidths B_T according to (6).

In Fig. 5, theoretical posterior bounds are shown for RSS measurements, utilizing 1 or 10 s measurement intervals, while considering SNR-based measurement selection and the proposed FIM-based measurement selection. As expected, the FIM-based measurement selection provides the best positioning

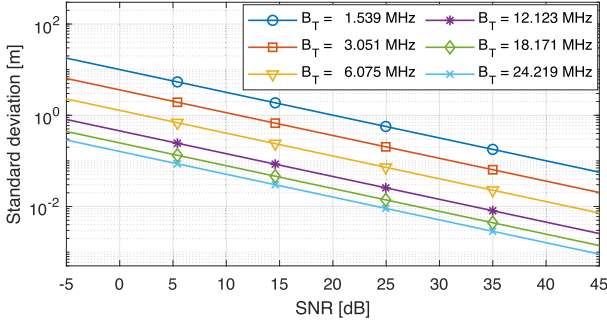


Fig. 4. CRLBs for ToA-based range estimation as a function of SNR with the considered DECT-2020 NR nominal bandwidths.

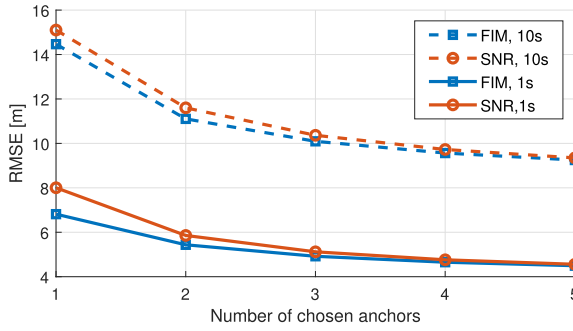


Fig. 5. Theoretical posterior bounds with RSS measurements for 1 and 10 s measurement intervals as a function of number of chosen anchors using SNR-based and FIM-based measurement selection approach.

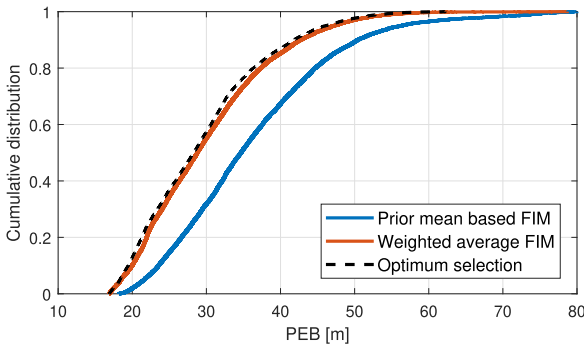


Fig. 6. Cumulative distribution of PEB for two FIM-based anchor selection method for a static example scenario presented in Section III-B.

performance. Nonetheless, when the number of used anchors is increased, the difference between the two anchor selection methods reduces, as the most important anchors are more likely to be included in the used measurement set. PEBs with range-based measurements are later shown and compared with EKF-based results in Section V-C.

The performance of the proposed particle filter enabled measurement selection approach utilizing weighted average FIMs, as presented in Section IV-D, is illustrated in Fig. 6 by considering a static scenario presented earlier in Fig. 1. Here, we emphasize that the static scenario used in evaluating results in Fig. 6 differs from the mobile scenario with multiple user tracks and anchor

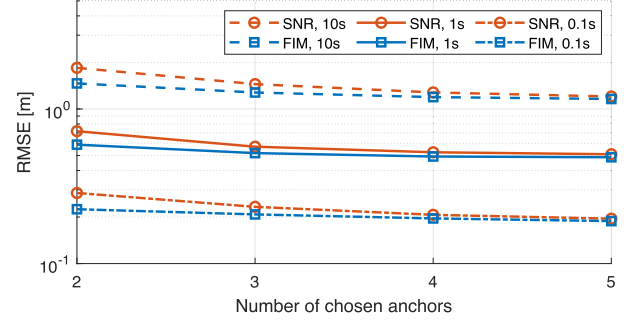


Fig. 7. RMSE as a function of number of chosen anchors for the proposed EKF using FIM- and SNR-based measurement selection with AoA measurements.

deployments used to evaluate the remaining positioning results in the article. Nonetheless, for the static scenario the prior covariance is defined as a diagonal matrix with standard deviations of 20 and 80 m for the x and y components, respectively. The cumulative PEB curves in Fig. 6 show that the weighted average FIM-based method outperforms the basic FIM method utilizing only the prior mean in FIM evaluations. The method utilizing the weighted average FIM over a larger area takes better into account the uncertainty in the predicted user position, thus enabling more accurate FIM evaluations at the cost of increased computational complexity.

C. Numerical Results for EKF Using AoA and Range Measurements

In Fig. 7, the RMSE as a function of the number of chosen anchors with AoA-based positioning is shown for the proposed FIM-based anchor selection and the baseline SNR-based method, while considering different measurement time intervals. The results show that the proposed FIM-based anchor selection method improves positioning accuracy compared to the SNR-based selection. For example, when utilizing two anchors with 1 s tracking interval, positioning error RMSE is 18.2% smaller with FIM-based (0.59 m) compared to SNR-based (0.72 m) anchor selection. The difference in performance between the FIM- and SNR-based methods decreases when the number of used anchors increases. This is intuitive, as with increased number of anchors there is a diminishing probability for missing the most important measurements, whereas with small number of anchors the selection is more critical.

Corresponding results for positioning with ToA-based ranging are presented in Fig. 8. Results show similar trend as with the AoA measurements, but the proposed FIM-based method has relatively better performance compared to the SNR-based method. For example, with two selected anchors and 1 s tracking interval, RMSE is 47.5% smaller with the FIM-based method (0.70 m) than with the SNR-based method (1.33 m).

For enabling more elaborate comparison between the two anchor selection methods, Table I presents the percentage change of positioning RMSEs for using the proposed FIM-based selection method with respect to the SNR-based method. In the

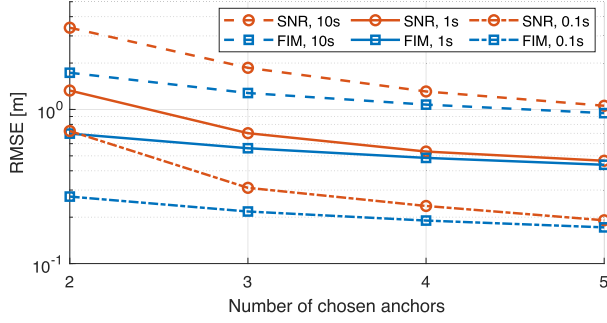


Fig. 8. RMSE as a function of number of chosen anchors for the proposed EKF using FIM- and SNR-based measurement selection with range measurements.

TABLE I

PERCENTAGE CHANGE OF POSITIONING ERROR RMSES BETWEEN THE TWO ANCHOR SELECTION METHODS, FIM-BASED VERSUS SNR-BASED METHOD

	AoA			Range		
	0.1 s	1 s	10 s	0.1 s	1 s	10 s
2	-21.5%	-18.2%	-20.7%	-62.5%	-47.5%	-49.0%
3	-10.7%	-9.1%	-11.7%	-29.8%	-20.1%	-31.4%
4	-5.2%	-6.1%	-6.7%	-19.6%	-9.1%	-17.8%
5	-3.5%	-4.4%	-3.6%	-10.0%	-5.8%	-10.4%

TABLE II

99.9TH PERCENTILES OF POSITIONING ERRORS IN METERS USING EKF TRACKING WITH TWO SELECTED ANCHORS

	AoA		Range	
	FIM	SNR	FIM	SNR
0.1 s	1.1266	1.6060	0.8422	6.1330
1 s	3.2499	4.4762	2.5995	16.9349
10 s	10.3525	14.8658	12.0734	35.4046

case of AoA measurements and two selected anchors, positioning RMSE is approximately 20% smaller with FIM-based selection compared to SNR-based selection, and approximately 10% smaller with three selected anchors. Moreover, the relative performance improvement of the FIM-based selection is further enhanced with range measurements, where with two selected anchors the reduction in positioning RMSE is up to 62.5%.

In certain use cases it is important to avoid large errors and provide consistent positioning performance over time. For this purpose, Table II shows the 99.9th percentile of positioning errors for the FIM- and SNR-based anchor selection methods by utilizing two anchors, and considering both AoA and range measurements as well as different tracking intervals. The results indicate that in terms of 99.9th percentile of positioning errors the proposed FIM-based selection method clearly outperforms the baseline SNR-based method with both AoA- and ToA-based measurements. For example, with range measurements and 1 s tracking interval, the 99.9th percentile error for the FIM-based and SNR-based selection method are 0.84 and 6.13 m, respectively, indicating approximately 86% error reduction with the proposed approach.

Finally, in Fig. 9, the RMSE as a function of time instant, considering 1 s measurement interval and two selected anchors,

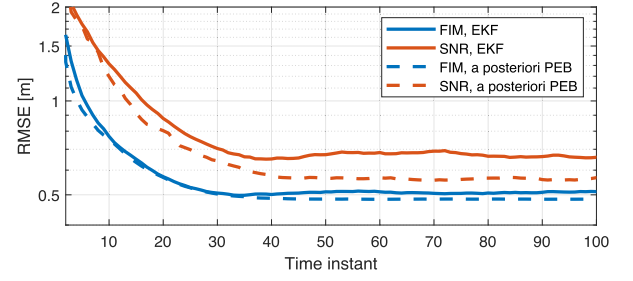


Fig. 9. RMSE as a function of time instants for FIM- and SNR-based measurement selection, considering EKF-based estimation and corresponding a posteriori bounds using range measurements.

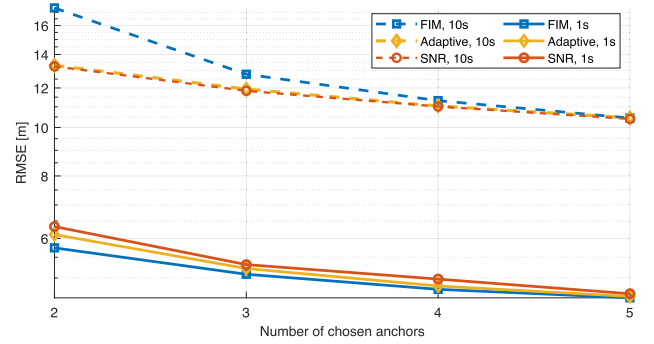


Fig. 10. Positioning error RMSE as a function of number of chosen anchors for the proposed particle filter using FIM, SNR, and adaptive FIM/SNR-based measurement selections with RSS measurements.

is illustrated for the EKF-based tracking, as well as for the a posteriori PEB, evaluated according to (17). In the figure, filtering gain stemming from the prior information obtained from the mobility model is seen as decreasing RMSE trend during the first time steps. It can be seen that the used EKF-based tracking approach outperforms the SNR-based benchmark method, while achieving close to PEB performance.

D. Numerical Results for Particle Filter Using RSS Measurements

In Fig. 10, the positioning error RMSE as a function of the number of chosen anchors with RSS measurements are presented for the proposed FIM-based selection and the SNR-based reference method, while considering measurement time intervals 1 and 10 s. In addition, the corresponding results for adaptive selection method, where the FIM- and SNR-based selection alternate according to prior estimate accuracy, are included. The results show that with a 1 s tracking interval, FIM-based method achieves better positioning accuracy than SNR-based method. The accuracy of the adaptive method falls between the accuracies of the FIM-based and SNR-based selection methods. For example, with two chosen anchors, the RMSE is 5.74 m for FIM-based selection, 6.11 m for adaptive selection, and 6.34 m for SNR-based selection. However, with the 10 s tracking interval the SNR-based anchor selection method performs better than the FIM-based selection, especially if the number of chosen anchors is 2 or 3. This is due to low accuracy prior information, leading to misleading FIM evaluations, while the SNR-based

TABLE III

PERCENTAGE CHANGE OF POSITIONING ERROR STATISTICS BETWEEN THE TWO ANCHOR SELECTION METHODS, FIM-BASED VERSUS SNR-BASED METHOD, USING RSS MEASUREMENTS AND PARTICLE FILTER TRACKING

	RMSE		95th percentile		99th percentile	
	1 s	10 s	1 s	10 s	1 s	10 s
2	-9.4%	+30.9%	-10.6%	+24.6%	-14.9%	+47.2%
3	-4.4%	+8.0%	-5.2%	+4.6%	-8.0%	+6.5%
4	-4.6%	+2.7%	-3.6%	0.0%	-5.7%	+2.0%
5	-1.9%	+0.4%	-1.6%	-0.9%	-3.1%	+0.5%

TABLE IV

95TH, 99TH, AND 99.9TH PERCENTILES OF POSITIONING ERRORS IN METERS USING PARTICLE FILTER TRACKING WITH TWO SELECTED ANCHORS WITH RSS MEASUREMENTS AND PARTICLE FILTER TRACKING

	1 s			10 s		
	95th	99th	99.9th	95th	99th	99.9th
FIM	10.875	15.339	22.787	32.139	56.275	120.129
SNR	12.159	18.025	29.815	25.799	38.240	56.622
Adaptive	11.677	17.328	27.721	25.818	38.453	57.723

method tends to inherently find good RSS measurements by emphasizing measurements in proximity to anchors. The adaptive method, instead, is able to avoid the largest errors, leading to similar positioning accuracy with the SNR-based selection method for all considered number of chosen anchors.

Table III presents the percentage change of positioning error statistics, considering RMSE, 95th percentile, and 99th percentile, for the FIM-based anchor selection method with respect to the SNR-based method when using RSS measurements and particle filter. It can be seen that the FIM-based anchor selection results in reduced positioning error compared to SNR-based method, when 1 s tracking interval is used, and conversely increased positioning error with 10 s tracking interval. However, when five chosen anchors and 10 s tracking interval are used, the relative difference is under 1% when comparing FIM-based and SNR-based anchor selection methods for all metrics. With the 1 s tracking interval, the accuracy of prior information increases, and consequently the proposed FIM-based method outperforms the baseline SNR-method. For example, using two chosen anchors, RMSE and 99th percentile are 9.4% and 14.9% smaller with FIM-based selection compared to SNR-based selection, respectively.

In Table IV, the 95th percentile, 99th percentile, and 99.9th percentile of positioning errors by utilizing two anchors are presented for FIM-based, SNR-based, and adaptive anchor selection methods, while considering RSS measurements and tracking intervals 1 and 10 s. As expected, with 1 s tracking interval the FIM-based selection outperforms the baseline SNR-based method in all three percentile scores. For example, 99.9th percentile error for the FIM-based and SNR-based selection method are 22.787 and 29.815 m, respectively, indicating roughly 24% error reduction for the FIM-based method. Again, with 10 s tracking interval, the performance of the FIM-based selection decreases, but with the adaptive anchor selection approach, the performance is comparable with the baseline signal-to-noise ratio (SNR)-based method.

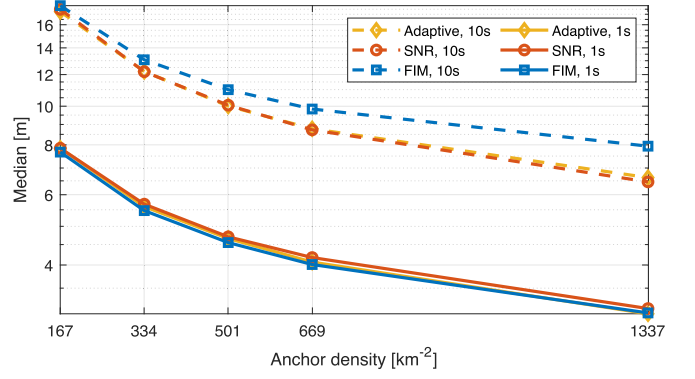


Fig. 11. Median as a function of anchor densities for the proposed particle filter using FIM, SNR, and adaptive FIM/SNR-based measurement selection with RSS measurements.

Finally, the effect of anchor density on achieved positioning accuracy is studied in Fig. 11, showing median positioning errors as a function of anchor density for different anchor selection approaches and measurement intervals. According to intuition, the results show that positioning error reduces with increased anchor density due to growing probability of obtaining high quality measurements. Considering different anchor selection methods, the results seem consistent with the earlier results. Again, when considering the 10 s measurement interval with RSS measurements, occasional large errors in the predicted user position limit the performance of the FIM-based selection method. However, by reducing the measurement interval to 1 s, the enhanced quality of prior information boosts the FIM-based selection method to reach the best performance, even in the challenging RSS-based positioning scenario.

VI. CONCLUSION

In this article, we extended our earlier research work considering positioning in IoT systems with a specific focus on the DECT-2020 NR standard. We utilized the proposed method of selecting the most important positioning measurements to avoid processing of negligible measurements, therefore enabling reduced energy consumption, latency, and computational load. Moreover, by exploiting measurement-based Fisher information from potential anchor measurements, evaluated through the means of prediction step of a tracking filter and associated mobility model, the proposed measurement selection method were used to minimize the a posteriori PEB. In addition to EKF-based tracking, we formulated a particle filter-based solution for challenging RSS-based positioning scenarios, and proposed a novel method to utilize tracked particles for FIM-based measurement selection.

Based on extensive set of numerical results considering specific DECT-2020 NR system parameters, the performance of the proposed measurement selection approach was assessed and compared against the baseline SNR-based selection scheme considering range, angle, and RSS measurements, while also addressing the corresponding theoretical bounds. In addition, up to a certain point the results can be applied to scenarios

with other wireless technologies, such as cellular 5G NR, when considering similar configurations of transmission bandwidth and antenna arrays. The presented results show that the proposed method outperforms the SNR-based solution, especially with range measurements, and a small number of used measurements. However, if the uncertainty of predicted user state is high, for example, when using long measurement intervals and relatively inaccurate RSS measurements, it is possible to utilize the proposed particle filter-based measurement selection with weighted average FIM, or alternatively the computationally low adaptive selection scheme. Furthermore, the proposed FIM-based measurement selection method significantly reduces the largest positioning errors (up to 60%), which can turn out to be a crucial asset for various IoT use cases and applications.

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